2. Particular Integral (y_p): Exponential Case

After finding the Complementary Function (y_c), the next step is to find the **Particular Integral (y_p)**. This part of the solution depends entirely on the function $F(x)$ on the right-hand side of the differential equation.

This file covers **Method 1**, which is used when $F(x)$ is an exponential function of the form e^{ax}.

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1. The Method

Given a differential equation $f(D)y = e^{ax}$:

The Particular Integral is written as:

$$y_p = \frac{1}{f(D)}e^{ax}$$

1.1 The Rule ($D \to a$)

To solve this, simply replace the differential operator D with the coefficient a from the exponent.

$$y_p = \frac{1}{f(a)}e^{ax}, \quad \text{provided } f(a) \neq 0$$

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2. The Case of Failure ($f(a) = 0$)

If replacing D with a results in a zero in the denominator ($f(a) = 0$), the method fails. To fix this, we apply the following rule:

1. Multiply the numerator by x.
2. Differentiate the denominator with respect to D ($f'(D)$).
3. Try substituting $D = a$ again.

$$y_p = x \cdot \frac{1}{f'(D)}e^{ax}$$

If the denominator is still zero, repeat the process (multiply by x again to get x^2, and differentiate the denominator again to get $f''(D)$).

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3. Solved Questions

Here are step-by-step examples from the lecture notes, covering both the standard case and the case of failure.

Question 1: Standard Case & Constants

Solve for y_p: $(D^2 - 6D + 9)y = 6e^{3x} + 7e^{-2x} - \ln 2$

Solution: We can split this into three separate parts for linearity:

$$y_p = y_{p1} + y_{p2} + y_{p3}$$

$$y_p = \frac{1}{(D-3)^2}(6e^{3x}) + \frac{1}{(D-3)^2}(7e^{-2x}) - \frac{1}{(D-3)^2}(\ln 2)$$

Part 1: $6e^{3x}$ **($a = 3$)** Substitute $D = 3$ into $(D-3)^2$: $(3-3)^2 = 0$. **Case of Failure!** * Multiply by x, differentiate denominator $2(D-3)$:

$$x \cdot \frac{1}{2(D-3)} 6e^{3x}$$

* Substitute $D = 3$ again: $2(3-3) = 0$. **Failure again!** * Multiply by x again (x^2), differentiate denominator again (2):

$$x^2 \cdot \frac{1}{2} 6e^{3x} = 3x^2 e^{3x}$$

Part 2: $7e^{-2x}$ **($a = -2$)** Substitute $D = -2$:

$$\frac{1}{(-2-3)^2} 7e^{-2x} = \frac{1}{(-5)^2} 7e^{-2x} = \frac{7}{25} e^{-2x}$$

Part 3: Constant Term $\ln 2$ **($a = 0$)** A constant can be written as $(\ln 2) \cdot e^{0x}$. Here, $a = 0$. Substitute $D = 0$:

$$\frac{1}{(0-3)^2} (\ln 2) = \frac{1}{9} \ln 2$$

Total y_p:

$$y_p = 3x^2 e^{3x} + \frac{7}{25} e^{-2x} - \frac{\ln 2}{9}$$

Question 2: Higher Order Failure

Solve for y_p: $(D^3 - 3D^2 + 3D - 1)y = e^x$

Solution: 1. **Identify Form:** The operator $(D^3 - 3D^2 + 3D - 1)$ is the expansion of $(D-1)^3$.

$$y_p = \frac{1}{(D-1)^3} e^x$$

2. **Apply Rule ($D \to 1$):** Substitute $D = 1$: $(1-1)^3 = 0$. **Case of Failure.** 3. **Fix 1:** Multiply by x, differentiate denominator ($3(D-1)^2$).

$$x \cdot \frac{1}{3(D-1)^2} e^x$$

Substitute $D = 1$: Still 0. 4. **Fix 2:** Multiply by x again (x^2), differentiate denominator ($6(D-1)$).

$$x^2 \cdot \frac{1}{6(D-1)} e^x$$

Substitute $D = 1$: Still 0. 5. **Fix 3:** Multiply by x again (x^3), differentiate denominator (6).

$$x^3 \cdot \frac{1}{6} e^x$$

Now the denominator is constant (6), which is not zero.

Final Answer:

$$y_p = \frac{x^3 e^x}{6}$$

Next Method: What if $F(x)$ is not exponential, but a polynomial like x^2 or x^3? Proceed to Method 2: Algebraic Case.